Mathematics · Practice Test 2

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Which list is ordered from **least to greatest**?

A. 0.72, ¾, √0.6, 78% B. ¾, 0.72, 78%, √0.6 C. 78%, √0.6, ¾, 0.72 D. √0.6, 0.72, ¾, 78%

2. Between which two whole numbers does √90 lie? A. 8 and 9 B. 9 and 10 C. 44 and 46 D. 89 and 91

3. *(Griddable)* Write 0.00045 in scientific notation. ☐

4. Which number is **rational**? A. √8 B. √16 C. π D. √2

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (−2, 1) and (2, 9)? A. 1 B. 2 C. 4 D. 8

6. A proportional relationship passes through the origin and the point (5, 15). What is the unit rate? A. 3 B. 5 C. 10 D. 15

7. *(Equation editor)* A candle is 20 cm tall and burns down 2 cm each hour. Write an equation for its height y after x hours. ☐

8. Which relation is a **function**?

A. {(2, 1), (2, 3), (4, 5)} B. {(1, 6), (2, 6), (3, 7)} C. {(5, 2), (5, 4), (6, 8)} D. {(0, 0), (0, 1), (1, 2)}

9. Which equation represents a **non-proportional** relationship? A. y = 3x B. y = 7x C. y = 3x + 4 D. y = ½x

10. *(Griddable)* For a direct variation, y = 36 when x = 4. What is y when x = 7? ☐

11. Which table shows a **proportional** relationship?

A. (1, 2)(2, 5)(3, 8) B. (1, 4)(2, 8)(3, 12) C. (1, 3)(2, 5)(3, 7) D. (2, 7)(3, 9)(4, 11)

12. *(Griddable)* Solve for x: 7x − 5 = 3x + 19. ☐

13. "Five more than three times a number equals twice the number plus twelve." Which equation models this? A. 3x + 5 = 2x + 12 B. 5x + 3 = 12x + 2 C. 3x − 5 = 2x + 12 D. 5 + 3 = 2x + 12

14. Two lines intersect at (−3, 2). What is the solution of the system? A. x = 2, y = −3 B. x = −3, y = 2 C. x = −1 D. No solution

15. On a line, right triangles are drawn to two different pairs of points. What must be true of the triangles? A. Congruent B. **Similar**, so the slope is the same C. Equal area D. Isosceles

16. Which situation represents a **proportional** relationship?

A. A plumber charges $50 plus $40 per hour B. A runner travels 6 miles per hour at a steady pace C. A tank has 30 L and loses 5 L per hour D. A gym costs $25 to join plus $5 per class

17. Which situation can be modeled by $4x + 6 = 2x + 18$?

A. Two accounts start at $6 and $18 and grow $4 and $2 per week; when are they equal? B. The product of 4 and 6 C. A rectangle's perimeter D. 4 plus 6 plus 18

18. A taxi charges $4 to start plus $3 per mile. Which equation gives the cost y for x miles? A. $y = 4x + 3$ B. $y = 3x + 4$ C. $y = 7x$ D. $y = 3x - 4$

19. *(Select TWO)* Which statements about the line $y = -3x + 6$ are true? A. The slope is −3 B. The slope is 6 C. The y-intercept is 6 D. The y-intercept is −3

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 5 and height 6. What is its volume? ($\pi \approx 3.14$) A. 150 B. 235.5 C. 471 D. 942

21. A sphere has radius 3. What is its volume? ($\pi \approx 3.14$) A. 28.26 B. 84.78 C. 113.04 D. 339.12

22. A cylinder has radius 2 and height 10. What is its **total** surface area? ($\pi \approx 3.14$) A. 125.6 B. 150.72 C. 175.84 D. 251.2

23. *(Griddable)* A right triangle has legs of 9 and 40. What is the length of the hypotenuse? ☐

24. What is the distance between (0, 0) and (6, 8)? A. 7 B. 10 C. 14 D. 48

25. The point (4, −3) is dilated by a scale factor of 2 about the origin. What is the image? A. (6, −1) B. (8, −6) C. (2, −1.5) D. (8, −3)

26. The point (−6, 2) is reflected over the x-axis. What is the image? A. (6, 2) B. (−6, −2) C. (6, −2) D. (2, −6)

27. The point (4, 1) is rotated 180° about the origin. What is the image? A. (−4, −1) B. (−1, 4) C. (1, −4) D. (4, −1)

28. Which transformation **preserves** congruence (keeps the same size)? A. Dilation by factor 2 B. Dilation by factor ½ C. Translation D. Dilation by factor 3

29. Two parallel lines are cut by a transversal. One angle measures 65°. What is its corresponding angle? A. 25° B. 65° C. 115° D. 180°

30. Two similar triangles have corresponding sides in a ratio of 3 : 4. A side of the smaller is 9. What is the matching side of the larger? A. 10 B. 11 C. 12 D. 13

31. The Pythagorean theorem relates the sides of a right triangle by which equation? A. $a + b = c$ B. $a^2 + b^2 = c^2$ C. $a^2 - b^2 = c^2$ D. $2a + 2b = c$

32. A rectangle measures 5 by 3. It is dilated by a scale factor of 2. What is the area of the image? A. 30 B. 45 C. 60 D. 90

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 4, 8, 10, 14. ☐

34. A trend line is modeled by y = 3x + 5. Predict *y* when x = 6. A. 18 B. 20 C. 23 D. 30

35. A scatterplot shows points with no clear pattern or direction. What association is shown? A. Positive linear B. Negative linear C. No association D. Strong linear

36. Which description suggests a **non-linear** association? A. A straight upward line B. A straight downward line C. A curved, U-shaped path D. Points exactly on one line

37. *(Griddable)* Find the simple interest on $1,200 at 6% for 3 years. ☐

38. For the same loan amount and length, how does a **higher** interest rate affect the total cost? A. Lowers it B. Raises it C. No effect D. Cancels the loan

39. Which plan is likely to grow the **most** over time? A. Saving nothing B. Saving a little, once C. Saving small amounts regularly with compound interest D. Spending it all

40. College costs $16,000. A family expects to contribute $6,000 and wants to save the rest over 20 months. How much must they save each month? A. $400 B. $500 C. $600 D. $800

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.